Deep Learning-Based Segmentation and Classification of Breast Lesions in Ultrasound Datasets

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Abstract

Breast cancer remains one of the leading causes of cancer death among women worldwide, and early detection through ultrasound imaging is a key factor in improving survival outcomes. Manual interpretation of breast ultrasound images, however, depends heavily on the availability of trained radiologists and is prone to subjectivity and inconsistency. This study investigates the use of YOLO26 for automated breast lesion detection, segmentation, and classification from ultrasound images. The study initially considered the Breast Ultrasound Images (BUSI) dataset and was subsequently expanded by incorporating images from the BUS-UCLM and BUS-BRA datasets to improve dataset diversity. The final balanced dataset contained 458 images per class across the Normal, Benign, and Malignant categories, with approximately 3.2k images after augmentation. YOLO26s-seg models were trained using a 70:20:10 training-validation-testing split and evaluated using bounding-box and segmentation mean Average Precision (mAP) metrics. Multiple experiments investigated augmentation strategies, hyperparameters, dataset size, and model size. The best-performing YOLO26s configuration on the approximately 3.2k image dataset achieved a box mAP@50 of 0.84192, box mAP@50:95 of 0.59732, mask mAP@50 of 0.83808, and mask mAP@50:95 of 0.55686. Increasing the augmented dataset to approximately 10,000 images improved mAP@50 but degraded mAP@50:95, indicating that additional augmented samples aided coarse detection without improving strict localization or mask quality. These results demonstrate the potential of YOLO26 for integrated breast lesion analysis and highlight the importance of augmentation design, dataset diversity, model capacity, and external validation.

Keywords breast cancer, breast ultrasound, YOLO26, deep learning, image classification, Roboflow, BUSI dataset, computer-aided diagnosis

I. INTRODUCTION

Breast cancer remains one of the most significant global health concerns. According to the World Health Organization, an estimated 2.4 million women were diagnosed with breast cancer and approximately 694,000 deaths were attributed to the disease worldwide in 2024. Breast cancer was also the most commonly diagnosed cancer among women in 164 of 186 countries during the same year [1]. These figures underscore the continued importance of effective approaches for early detection, diagnosis, and treatment.

Medical imaging plays a central role in breast cancer assessment. Among available imaging modalities, ultrasound is particularly useful because it is non-invasive, does not expose patients to ionizing radiation, and can assist in differentiating breast tissue abnormalities. It is also commonly used as a complementary modality, particularly when breast tissue characteristics make the interpretation of other imaging modalities more difficult. Nevertheless, ultrasound images present several challenges for automated analysis. Speckle noise, low contrast, irregular lesion boundaries, differences in acquisition equipment, and variation in lesion appearance can make the localization and delineation of breast lesions difficult.

Deep learning has increasingly been applied to medical image analysis because of its ability to learn visual representations directly from imaging data. Convolutional neural networks and related architectures have been applied to breast ultrasound classification, detection, and segmentation. The Breast Ultrasound Images Dataset (BUSI), for example, contains ultrasound images belonging to normal, benign, and malignant categories and provides corresponding ground-truth information that can be used for machine learning research [2].

Object detection approaches such as the You Only Look Once (YOLO) family are particularly relevant to this problem because they perform localization and classification within a unified framework [5]. Modern segmentation-enabled YOLO models additionally generate pixel-level masks, allowing a system to provide not only the predicted lesion category and bounding region but also a delineation of the lesion itself.

The present study focuses on YOLO26 segmentation for breast ultrasound images. Unlike a purely classification-oriented approach, the proposed system is designed to produce three forms of output: the predicted class of the image or lesion, the bounding box locating the lesion, and a segmentation mask representing the lesion boundary. This integrated approach is intended to provide more diagnostic information than classification alone.

The initial version of this study was based primarily on a balanced subset of BUSI containing 458 images per class. As the project progressed, the dataset and experimental methodology were expanded. BUS-UCLM and BUS-BRA were incorporated alongside BUSI, resulting in a balanced dataset of 458 images per class. Multiple training experiments were subsequently conducted to examine the effects of augmentation, hyperparameters, dataset size, and model size.

The main objectives of this study are:

1. To develop a YOLO26-based model capable of detecting, segmenting, and classifying breast lesions in ultrasound images.
2. To construct a balanced breast ultrasound dataset using images from BUSI, BUS-UCLM, and BUS-BRA.
3. To investigate augmentation strategies appropriate for breast ultrasound images.
4. To evaluate different training configurations and dataset sizes.
5. To compare YOLO26-small and YOLO26-nano models in terms of detection and segmentation performance.
6. To compare the resulting performance with relevant previous studies involving breast ultrasound lesion detection and segmentation.
7. To identify remaining improvements necessary for future deployment and external validation.

II. RELATED WORK

A. Breast Ultrasound Datasets

The BUSI dataset introduced by Al-Dhabyani et al. contains breast ultrasound images categorized into normal, benign, and malignant cases [2]. The dataset was collected from 600 female patients between 25 and 75 years of age and originally contained 780 images. Ground-truth masks were also provided for lesion-containing images. Because the dataset supports classification, detection, and segmentation tasks, it has been widely used as a benchmark for breast ultrasound research.

The present study uses BUSI as its primary dataset source but expands it with BUS-UCLM and BUS-BRA. This expansion was motivated by the need to expose the model to a wider range of ultrasound appearances rather than relying exclusively on a single dataset.

BUS-UCLM contains 683 images obtained from 38 patients and includes benign, malignant, and normal categories. Arioua et al. used this dataset to investigate a localization-guided segmentation approach in which YOLOv11-n was first used to localize suspicious regions before a separate segmentation model was applied to the detected region [3]. Their results demonstrate the relevance of combining lesion localization with detailed segmentation for breast ultrasound analysis. The same study reported that DeepLabV3 achieved a Dice score of 84.49% and an IoU of 73.57% under its direct semantic segmentation approach [3].

BUS-BRA provides another source of breast ultrasound images and was included in the present study to increase dataset diversity. Of particular relevance, BUS-BRA serves as the source corpus in the cross-dataset benchmark by Diptho et al. [7] discussed in Section II.C, and it is one of the few public breast ultrasound datasets that provides patient identifiers suitable for leakage-controlled splitting. The present project therefore moved from a BUSI-centered dataset toward a multi-dataset configuration containing BUSI, BUS-UCLM, and BUS-BRA.

B. Deep Learning for Breast Lesion Segmentation

U-Net and its variants have traditionally been important architectures for biomedical image segmentation. Their encoder-decoder structure and skip connections allow them to preserve spatial information while learning increasingly abstract image features. Numerous studies have demonstrated strong segmentation performance using U-Net-based approaches on breast ultrasound images.

A recent study by Rahmawati and Siregar investigated the effect of augmentation-based preprocessing on U-Net breast lesion segmentation [4]. The study trained on the BrEaST dataset and evaluated generalization on the independent QAMEBI dataset. The researchers found that augmentation policies could substantially affect external-dataset performance, with carefully selected combinations improving the Jaccard Similarity Index and recall compared with a basic preprocessing pipeline. Their findings emphasize that augmentation should not be treated merely as a way of increasing dataset size; the type of transformation can influence whether a model learns useful variations or unrealistic image characteristics [4].

This issue is especially relevant to ultrasound images because transformations that are acceptable in natural-image datasets may not necessarily correspond to realistic medical-image variations. Consequently, augmentation strategies in the present study were progressively revised throughout the experiments.

C. YOLO-Based Breast Ultrasound Analysis

The YOLO family was originally introduced as a unified object-detection framework capable of predicting object locations and class probabilities directly from an image [5]. Later generations extended the framework and improved detection and segmentation capabilities.

Several recent studies have specifically investigated YOLO models for breast ultrasound analysis. Yu et al. benchmarked twelve YOLO variants from the YOLOv5, YOLOv8, YOLOv9, and YOLO11 families using BUSI and BUS-UCLM under both same-database and cross-database conditions [6]. Their results showed that YOLO variants could achieve strong same-dataset segmentation performance, while

performance decreased substantially under cross-dataset evaluation. This observation is particularly important for breast ultrasound because images from different institutions may differ in scanners, acquisition protocols, image quality, and lesion appearance. The same study also demonstrated that smaller YOLO models can remain competitive while requiring less computational resources, supporting the investigation of YOLO26-nano and YOLO26-small in the present project rather than assuming that increasing model size will automatically result in better performance.

Most closely related to the present work is the cross-dataset benchmark of Diptho et al. [7], which trained five lightweight YOLO instance-segmentation architectures (YOLOv8n, YOLO11n, YOLO11s, YOLO26n, and YOLO26s) exclusively on BUS-BRA under patient-grouped splitting and evaluated them without adaptation on BUS-UCLM. Their study assessed ultrasound-specific preprocessing, test-time augmentation (TTA), and Optuna-based hyperparameter optimization across 40 paired internal-external comparisons. Cross-dataset evaluation demonstrated consistent lesion delineation on BUS-UCLM, with matched Dice ranging from 0.861 to 0.875 and matched IoU from 0.764 to 0.786 across the five architectures. Preprocessing improved mask mAP@50-95 across all ten checkpoints on both datasets, while TTA improved detection-adjusted Dice despite having little effect on mAP@50-95. Optuna tuning improved external mAP@50-95 across all five architectures despite limited internal gains. The study also showed that internal ranking does not reliably predict external ranking, that the malignant class carries the larger cross-dataset penalty, and that a substantial share of external misses at the fixed operating threshold were sub-threshold detections rather than complete failures to localize. ONNX Runtime deployment achieved 5.65–13.62 FPS on CPU.

That study establishes the importance of external validation, detection-aware evaluation, and efficient deployment for lightweight YOLO breast ultrasound segmentation. It is, however, restricted to the two-class benign-versus-malignant task and trains on a single source corpus. The present study differs in three respects. First, it addresses the three-class Normal/Benign/Malignant setting required for screening-oriented analysis by combining BUSI, BUS-UCLM, and BUS-BRA into a balanced multi-source dataset. Second, it reports a detailed confusion-matrix analysis that separates benign-malignant misclassification from background-to-lesion errors—a distinction that aggregate mAP or Dice metrics do not expose. Third, it decomposes the mAP@50 versus mAP@50:95 discrepancy across dataset size, augmentation policy, and model capacity, whereas prior work has emphasized detection-adjusted overlap alone. The two studies are therefore complementary: [7] establishes cross-dataset transfer behavior for two-class YOLO segmentation, while the present work extends the analysis to three-class detection, segmentation, and classification within a multi-source training framework.

D. Dataset Augmentation and Generalization

Data augmentation is commonly used to reduce overfitting and increase the variety of training samples. However, augmentation must be selected carefully in medical imaging. The recent study by Rahmawati and Siregar found that different augmentation operators had different effects on external generalization [4]. Some transformations improved performance, whereas others produced inconsistent results when applied to external data.

This observation was reflected in the present project. Early experiments used relatively aggressive transformations, including vertical flipping, large rotations, strong shear, blur, and noise. These settings were subsequently revised because some transformations could produce ultrasound appearances that were less representative of realistic breast imaging conditions. Diptho et al. [7] similarly restricted their online geometric augmentation to conservative operations (rotation up to 8°, translation up to 5%, scaling up to 15%, shear up to 2°, and horizontal flipping) while deliberately disabling mosaic, mixup, cutmix, copy-paste, and random erasing, on the grounds that compositing multiple ultrasound frames

or occluding regions produces images with no physical counterpart in a real examination. The project therefore moved toward more controlled augmentation involving horizontal flipping, limited cropping, moderate rotation and shear, and controlled exposure changes. The augmentation policy was also modified across experiments to determine a more appropriate balance between diversity and preservation of anatomical appearance.

E. Research Gap

Existing research demonstrates that deep learning can achieve strong performance in breast ultrasound segmentation, but several issues remain. First, high performance on a single dataset does not necessarily imply strong generalization to images obtained from other datasets or institutions; Diptho et al. [7] quantified this effect for two-class YOLO segmentation, and Yu et al. [6] observed it across twelve YOLO variants. Second, augmentation strategies can improve or degrade generalization depending on the transformation [4]. Third, increasing dataset size or model capacity does not necessarily produce proportional improvements in stricter evaluation metrics. Finally, the three-class Normal/Benign/Malignant setting, which is closer to the screening use case than a two-class benign/malignant formulation, has received comparatively little attention in the YOLO-based breast ultrasound literature.

The present study addresses these issues by combining multiple breast ultrasound datasets and conducting controlled experiments involving augmentation, dataset size, hyperparameters, and model size. Rather than reporting only a single training run, the study examines how these factors affect YOLO26 detection and segmentation performance, and it extends the cross-dataset line of inquiry established by [7] to the three-class multi-source setting.

III. METHODOLOGY

A. Dataset Construction

The study used a combined breast ultrasound dataset assembled from three publicly available sources: the Breast Ultrasound Images Dataset (BUSI) [13], BUS-UCLM [14], and BUS-BRA. BUSI [15] served as the primary dataset, while BUS-UCLM and BUS-BRA were used to supplement classes with fewer available images and to expose the model to a wider range of ultrasound appearances than a single-source dataset would provide.

The final balanced dataset contained 458 images per class across three categories: Normal, Benign, and Malignant, for a total of 1,374 original images. Images were annotated using Roboflow and prepared for YOLO26 instance segmentation. Auto-orientation was applied, and all images were resized to 640 × 640 pixels to provide a consistent input size during training. The dataset was divided into training, validation, and testing subsets using a 70:20:10 ratio.

To increase the effective size and variety of the training set, augmentation was applied using different multipliers, producing three augmented configurations of approximately 3.2k, 5.1k, and 9.8k images. These correspond approximately to 3×, 5×, and 10× augmentation of the original balanced dataset. The augmentation policy was revised during the study (see Section III-C) to emphasize transformations considered representative of realistic breast ultrasound variation. Roboflow was chosen for this stage because it provides a straightforward labeling interface, keeps annotations organized by class and project version, and can export the labeled dataset directly in a YOLO-compatible format, which simplifies the transition from annotation to model training. As of this stage of the project, annotation of the 458 images per class has been completed and organized within the Roboflow workspace, forming the labeled dataset that will be used for model training.

B. Model Architecture

The study used the YOLO26 instance-segmentation family. YOLO26, introduced by Ultralytics in September 2025, is designed around three principles: simplicity, efficiency, and innovation. [8] Unlike earlier YOLO generations that relied on Non-Maximum Suppression (NMS) for post-processing, YOLO26 eliminates NMS entirely, enabling native end-to-end predictions that reduce inference latency and simplify deployment. [8][9] It also removes Distribution Focal Loss (DFL), which had complicated model export in prior versions, improving compatibility with ONNX, TensorRT, and TFLite. [2] For training, YOLO26 introduces ProgLoss (progressive loss balancing) and STAL (small-target-aware label assignment), which improve convergence stability and detection of small or occluded objects. [2][8] These characteristics are relevant to breast ultrasound analysis, where lesions may be small and subtle. The segmentation-enabled variant used in this study retains YOLO's single-stage pipeline while adding mask prediction heads that generate pixel-level delineations alongside bounding boxes and class predictions. [15]

Two model variants were evaluated:

- YOLO26s-seg (small), with approximately 10.4 million parameters
- YOLO26n-seg (nano), with approximately 2.7 million parameters

Both models were trained at an input resolution of 640 × 640 pixels. The segmentation-enabled variants were selected because the study required three forms of output: the predicted class, a bounding box locating the lesion, and a pixel-level mask delineating the lesion boundary. The two model sizes were chosen to examine whether a lighter model could achieve competitive performance while reducing computational requirements.

C. Augmentation Strategy

Data augmentation was treated as a methodological component requiring explicit control rather than a generic means of increasing dataset size. Early experiments used an intentionally broad augmentation policy that included vertical flipping, rotations up to ±45°, large cropping ranges, shear, blur, and added noise. This policy was subsequently revised because several of these transformations could generate ultrasound appearances that were not representative of realistic breast imaging conditions — for example, vertical flips and large rotations can place anatomical structures in orientations that would not occur in practice.

TABLE I. Revised Augmentation Parameters

Transformation	**Setting**
Horizontal flip	Applied
Crop	0% min. zoom, 10% max. zoom
Rotation	−15° to +15°
Shear	±10° horizontal, ±10° vertical
Brightness	−10% to +10%
Exposure	−10% to +10%

Vertical flipping, large rotations, blur, and added noise were removed from the pipeline. The retained transformations were selected to reflect plausible variation in breast ultrasound acquisition — horizontal flipping accounts for left/right breast symmetry, limited cropping and small rotations reflect realistic probe positioning variation, moderate shear approximates soft-tissue deformation, and bounded brightness and exposure adjustments approximate scanner gain and display variation. This revision was informed by recent work showing that augmentation operators do not have uniform effects on external generalization in breast ultrasound segmentation [4], and that matching augmentation to the characteristics of the imaging modality is more important than maximizing the number or intensity of transformations.

D. Training Procedure

Experiments were primarily conducted in Google Colab using an NVIDIA L4 GPU. The training procedure was developed iteratively across three stages.

The methodology described here reflects the final stabilized configuration used for the dataset-size and model-size comparisons; the evolution that led to this configuration is presented in Section IV-A.

The initial learning rate of 0.001 was selected after preliminary experiments showed that higher learning rates produced unstable training, including exploding gradients and unstable loss behavior. Weight decay and cosine scheduling were introduced to improve numerical stability and control overfitting. Early stopping was used to terminate training when validation performance no longer improved, preventing unnecessary computation and reducing overfitting risk.

The same general training configuration was maintained across the dataset-size experiments so that the effect of dataset size could be compared consistently. After the dataset-size comparison, the best-performing configuration was used for the model-size comparison between YOLO26s-seg and YOLO26n-seg.

E. Evaluation Metrics

Model performance was evaluated using standard object detection and instance-segmentation metrics:

- Box mAP@50 — mean Average Precision for bounding boxes at an IoU threshold of 0.50
- Box mAP@50:95 — mean Average Precision for bounding boxes averaged across IoU thresholds from 0.50 to 0.95 in steps of 0.05
- Mask mAP@50 — mean Average Precision for segmentation masks at an IoU threshold of 0.50
- Mask mAP@50:95 — mean Average Precision for segmentation masks averaged across IoU thresholds from 0.50 to 0.95

Precision and recall were also recorded for bounding-box and mask predictions at the best-performing checkpoint. The distinction between mAP@50 and mAP@50:95 was treated as particularly important: mAP@50 measures whether a lesion is detected and broadly localized, while mAP@50:95 places greater weight on precise localization and segmentation quality. Reporting both metrics allowed the study to distinguish between models that detect lesions reliably and models that delineate them accurately.

IV. RESULTS AND DISCUSSION

A. Training Behavior and Configuration Evolution

The training procedure evolved across three stages. Stage 0 established a baseline using the original configuration; Stage 1 addressed augmentation realism and training stability; Stage 2 applied the stabilized configuration to the dataset-size and model-size experiments reported in Sections IV-B and IV-C.

Stage 0 — Baseline configuration.

The first stage consisted of two training runs using the initial training code and settings provided as the starting configuration. The optimizer was left on automatic selection, and no major modifications were made to the training configuration during these runs. The purpose of Stage 0 was to establish the baseline behavior of the model before applying team-developed modifications.

The first baseline run reached a box mAP@50 of approximately 0.837 and a mask mAP@50 of approximately 0.840, with corresponding mAP@50:95 values of approximately 0.575 and 0.524. Training stopped after 88 epochs because the early-stopping patience limit was reached. The second baseline run increased patience from 30 to 100 epochs, allowing training to continue to approximately 158 epochs; however, the additional training did not produce a higher peak performance, as the strongest validation results still occurred around epochs 57–58. Training and validation loss patterns also indicated that continued training increased the separation between

training and validation performance, suggesting that training duration had to be considered together with overfitting rather than treated as an independent method of improving accuracy.

Stage 1 — Augmentation realism and training stability.

The second stage focused on improving the realism of the image augmentations and addressing training instability. The original augmentation policy was considered too aggressive for breast ultrasound images: it included vertical flipping, large rotation ranges, substantial cropping, blur, and noise, all of which could produce ultrasound appearances that were not representative of realistic imaging conditions. The policy was therefore revised toward more conservative transformations. Vertical flipping, excessive rotation, blur, and added noise were removed or reduced, and the revised policy emphasized horizontal flipping, limited cropping, smaller rotations, and reduced shear.

The optimizer was also changed to AdamW in Run 3. This change alone, however, produced an excessively high effective learning rate and unstable training, including exploding gradients and unstable loss behavior. The experiment demonstrated that changing the optimizer without explicitly controlling the learning rate was insufficient. For Run 4, an explicit initial learning rate (LR0) of 0.001 and a weight decay of 0.0005 were introduced to improve numerical stability and reduce the occurrence of NaN losses while also helping control overfitting. Run 4 produced a substantial improvement over Run 3, reaching a box mAP@50:95 of 0.59117 and a mask mAP@50:95 of 0.54218. This represented the strongest result at that point in the experiment sequence.

Stage 2 — Final configuration.

After establishing a stable training configuration, the third stage focused on determining how dataset size and model size affected performance.

TABLE II. Final Training Configuration

Parameter	Value
Optimizer	AdamW
Initial learning rate (LR0)	0.001
Final learning-rate factor (LRF)	0.01
Learning-rate schedule	Cosine
Weight decay	0.001
Warm-up epochs	5
Batch size	32
Early-stopping patience	20 epochs
Input resolution	640 × 640

Cosine learning-rate scheduling, a final learning-rate factor of 0.01, five warm-up epochs, and a modified weight decay of 0.001 were introduced during Stage 2 to further stabilize training. The resulting configuration was then held relatively constant during the dataset-size comparison so that dataset size could be evaluated without repeatedly changing the major training variables. The augmentation policy described in Section III-C — horizontal flip, crop 0% minimum / 10% maximum, rotation ±15°, shear ±10°, brightness ±10%, exposure ±10% — was applied to generate the three augmented datasets used in Stage 2, corresponding to approximately 3.2k, 5.1k, and 9.8k images. The best-performing dataset configuration was subsequently selected for the model-size comparison between YOLO26s-seg and YOLO26n-seg.

Summary of training behavior.

Across all three stages, both models demonstrated a typical convergence pattern. Training losses decreased consistently throughout training, indicating that the models continued to fit the training

data. Validation losses decreased rapidly during the early epochs before reaching a relatively stable region with some fluctuation, and the precision, recall, and mAP metrics followed a similar pattern — a rapid initial rise followed by a gradual plateau. After the peak validation region, the metrics fluctuated rather than demonstrating sustained improvement. This behavior, together with the configured early-stopping patience, indicated that continued training beyond the convergence region provided limited generalization benefit and that the primary limitation was not simply insufficient training time.

B. Dataset-Size Experiments

The stabilized Stage 2 configuration was used to compare three augmented dataset sizes: approximately 3.2k, 5.1k, and 9.8k images, corresponding to 3×, 5×, and 10× augmentation of the original balanced dataset.

These three multipliers were selected to span the plausible range of useful augmentation for this dataset. The smallest augmented dataset was expected to be viable for segmentation training without severe overfitting; the 10× setting represented the practical upper bound given the available Colab GPU time budget; and the 5× setting provided a midpoint that would reveal whether the effect of dataset size was monotonic or plateaued.

TABLE III. Dataset-Size Comparison

Metric	**~3.2k Images**	**~5.1k Images**	**~9.8k Images**
Box mAP@50	0.84192	0.83578	**0.84863**
Box mAP@50:95	**0.59732**	0.56927	0.57283
Mask mAP@50	0.83802	0.82756	**0.84145**
Mask mAP@50:95	**0.55686**	0.52842	0.51877

The ~9.8k configuration achieved the highest box and mask mAP@50 values, indicating that the larger augmented dataset supported reliable coarse lesion detection. However, the ~3.2k configuration achieved the highest box and mask mAP@50:95 values, indicating stronger performance under stricter IoU thresholds. Because mAP@50:95 evaluates performance across progressively stricter overlap requirements, this difference indicates that the additional augmented images did not translate into more precise localization or segmentation boundaries. The ~3.2k configuration also required less computational time, with the progress report documenting approximately 30–45 seconds per epoch.

These results suggest that increasing the number of augmented images does not automatically improve localization and segmentation quality. Excessive augmentation may introduce additional image variations without providing new information useful for precise delineation. The quality and appropriateness of augmented samples therefore appears to be more important than the raw number of generated images. The ~3.2k configuration was selected for the subsequent model-size comparison because it produced the strongest strict mAP@50:95 values and required less computational time.

An important caveat applies to this comparison. The larger datasets were generated primarily through augmentation of the same underlying patient images rather than by adding thousands of genuinely new cases. Increasing the number of unique patients and imaging sources would be a different experiment and may improve generalization in ways that synthetic augmentation alone cannot reproduce.

C. Model-Size Comparison

The ~3.2k dataset was selected for the model-size comparison because it produced the strongest mAP@50:95 results in the dataset-size experiment while requiring the least computational time. Using ~3.2k kept the dataset configuration fixed at the setting that had already been shown to favor strict localization and segmentation quality, so that the

effect of model capacity could be examined without simultaneously introducing a dataset that had been observed to degrade mAP@50:95 performance.

TABLE IV. Model-Size Comparison

Metric	YOLO26n-seg	YOLO26s-seg
Box mAP@50	**0.84538**	0.84192
Box mAP@50:95	0.58225	**0.59732**
Mask mAP@50	**0.84406**	0.83802
Mask mAP@50:95	0.53342	**0.55686**

The two models achieved comparable performance at the 0.5 IoU threshold. YOLO26n-seg produced slightly higher box and mask mAP@50 values, while YOLO26s-seg produced higher mAP@50:95 values for both detection and segmentation. The difference between the two models was considerably more apparent under the stricter metric than under mAP@50.

This pattern suggests that the two models have different strengths. Both detect lesions reliably at the 0.5 IoU threshold, but YOLO26s-seg provides more accurate localization and segmentation when greater overlap with ground-truth annotations is required. The nano model achieved competitive performance under the less strict metric while using approximately 2.7 million parameters compared with approximately 10.4 million for the small model. Model selection therefore depends on whether the application places greater emphasis on computational efficiency or on strict localization and segmentation precision.

D. Classification and Localization Errors

To examine the types of errors produced by the models, normalized confusion matrices were analyzed for both experiments.

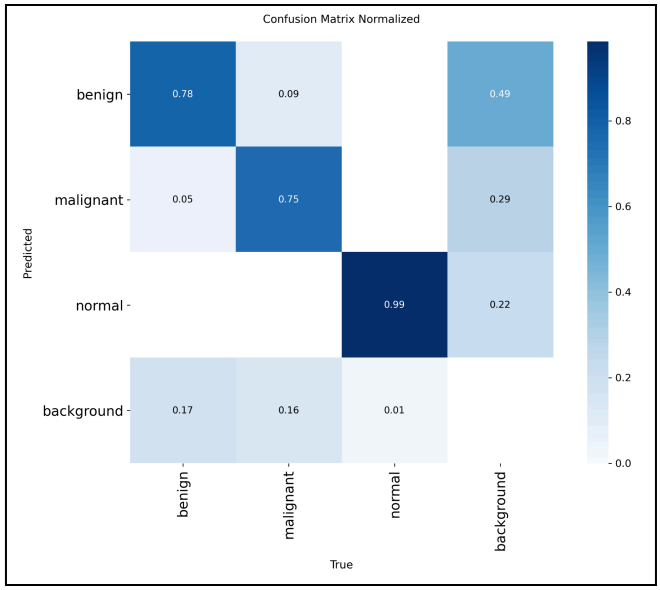


Fig. 3. Normalized confusion matrix of YOLO26n-seg.

The YOLO26n-seg model demonstrated relatively strong recognition of the normal class, with diagonal values of approximately 0.78 for Benign, 0.75 for Malignant, and 0.99 for Normal. Misclassification between benign and malignant cases was present but remained relatively limited: approximately 9% of true malignant samples were predicted as benign, and approximately 5% of true benign samples were predicted as malignant.

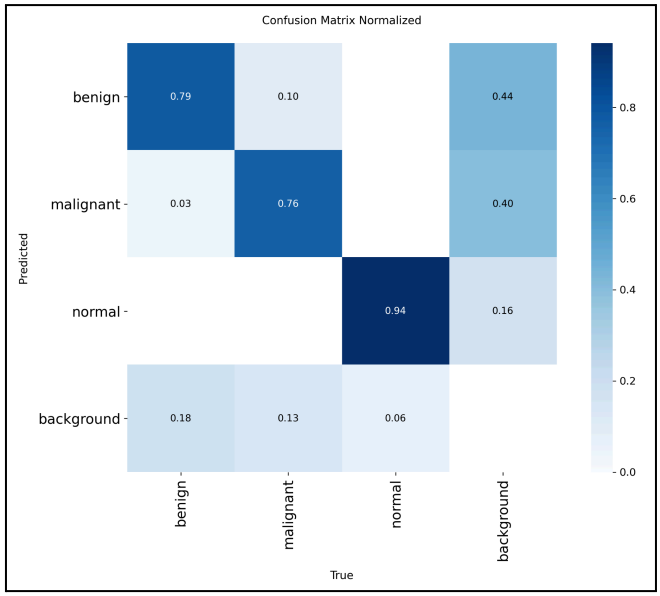


Fig. 4. Normalized confusion matrix of YOLO26s-seg.

A similar pattern was observed for YOLO26s-seg, with diagonal values of approximately 0.79 for Benign, 0.76 for Malignant, and 0.94 for Normal. Benign–malignant confusion remained limited: approximately 10% of true malignant samples were predicted as benign, and approximately 3% of true benign samples were predicted as malignant. The normal class remained highly recognizable, while benign and malignant cases showed greater uncertainty.

More prominent than benign–malignant confusion were background-related errors. In the YOLO26s-seg matrix, approximately 44% of the normalized background column was associated with benign predictions and 40% with malignant predictions. The YOLO26n-seg model showed a similar pattern, with approximately 49% and 29% of the normalized background column associated with benign and malignant predictions, respectively.

These errors should not be characterized simply as the model confusing benign and malignant lesions. Much of the error appears to originate from distinguishing actual lesion regions from surrounding background — a particularly challenging task in breast ultrasound, where texture and intensity patterns in normal tissue can resemble lesion boundaries. This suggests that separating lesion regions from visually similar background structures remains an important challenge for the models, potentially associated with the heterogeneous texture and low-contrast boundaries commonly present in breast ultrasound images.

E. Precision-Recall Analysis

Figures 5 and 6 show the precision-recall curves for bounding-box and mask predictions from both models. The curves provide class-specific Average Precision (AP) values that reveal how performance is distributed across the Normal, Benign, and Malignant categories.

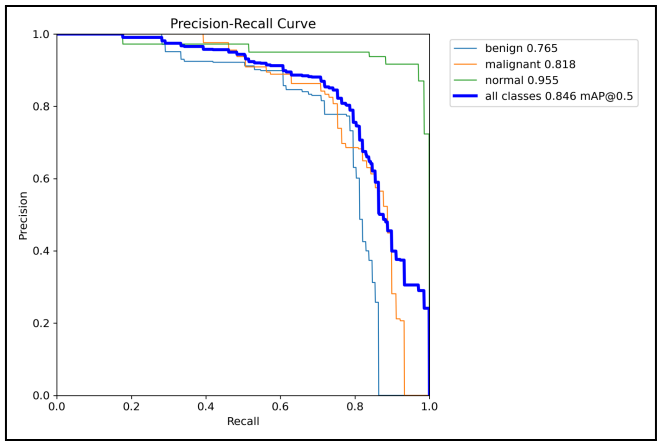


Fig. 5(a). Precision-recall curves for bounding-box detection YOLO26n-seg.

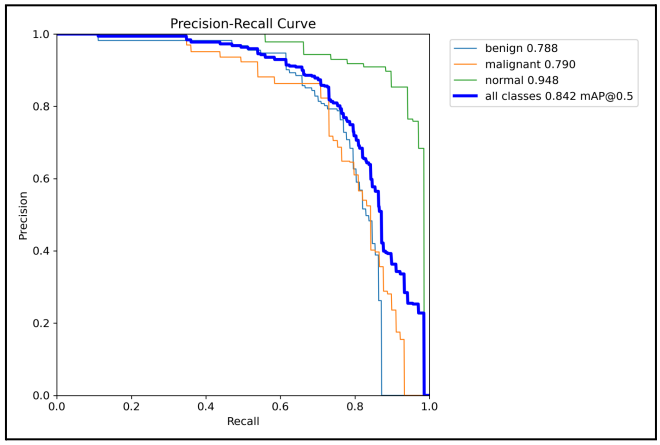


Fig. 5(b). Precision-recall curves for bounding-box detection YOLO26s-seg.

Figures 5(a) and 5(b) show the precision-recall curves for bounding-box detection from YOLO26n-seg and YOLO26s-seg, respectively. YOLO26n-seg achieved an overall mAP@0.5 of 0.846, with class-specific AP values of 0.765 for Benign, 0.818 for Malignant, and 0.955 for Normal. YOLO26s-seg achieved an overall mAP@0.5 of 0.842, with AP values of 0.788 for Benign, 0.790 for Malignant, and 0.948 for Normal.

In both panels the Normal class exhibited the strongest performance, maintaining high precision across a wider range of recall values than the other two classes. Precision for the Benign and Malignant classes declined more noticeably as recall increased, particularly beyond approximately 0.75–0.80,

indicating that increasing sensitivity for these classes results in more false-positive detections. At the class level, YOLO26n-seg achieved higher AP for Malignant (0.818 vs. 0.790) and Normal (0.955 vs. 0.948) cases, while YOLO26s-seg achieved higher AP for Benign cases (0.788 vs. 0.765). The overall difference between the two models at mAP@0.5 was small (0.846 vs. 0.842), and the class-level results move in different directions, indicating that the two models are broadly comparable in bounding-box precision-recall performance under the evaluated configuration.

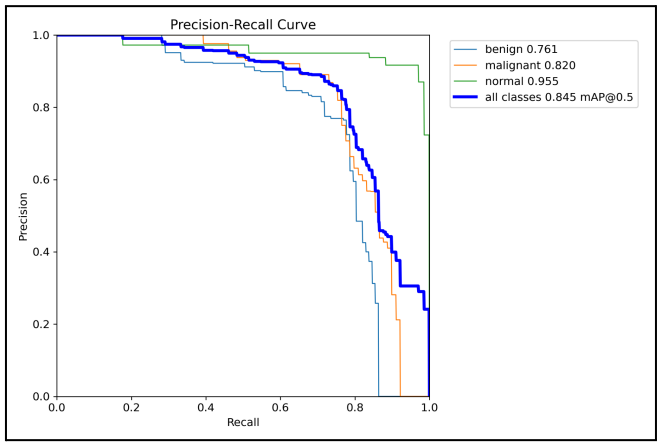


Fig. 6(a). Precision-recall curves for segmentation-mask prediction YOLO26n-seg.

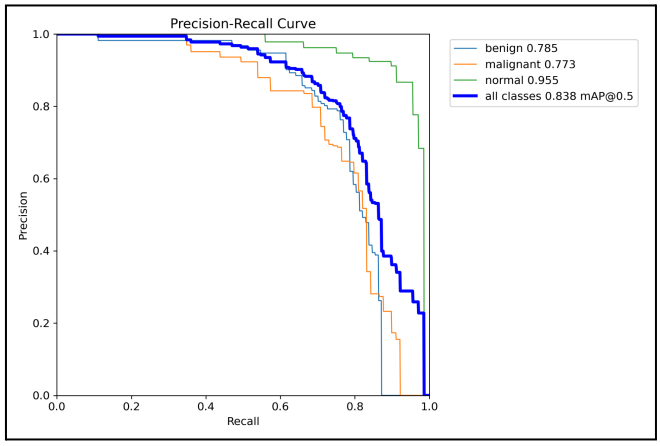


Fig. 6(b). Precision-recall curves for segmentation-mask prediction YOLO26s-seg.

YOLO26n-seg achieved an overall mask mAP@0.5 of 0.845, with class-specific AP values of 0.761 for Benign, 0.820 for Malignant, and 0.955 for Normal. YOLO26s-seg achieved an overall mask mAP@0.5 of 0.838, with AP values of 0.785 for Benign, 0.773 for Malignant, and 0.955 for Normal.

As in the bounding-box results, the Normal class maintained the highest and most stable precision across the recall range in both panels, while precision for the Benign and Malignant classes declined more sharply as recall increased. The Malignant class showed the lowest AP in the YOLO26s-seg panel (0.773), whereas the Benign class showed the lowest AP in the YOLO26n-seg panel (0.761). At the class level, YOLO26n-seg achieved higher AP for Malignant cases (0.820 vs. 0.773), while YOLO26s-seg retained higher AP for Benign cases (0.785 vs. 0.761). Both models produced equal AP for the Normal class (0.955).

Taken together with the bounding-box results, the mask precision-recall curves show a similar pattern: the two models are comparable in overall mask performance at the 0.5 IoU threshold (0.845 vs. 0.838), and the class-level differences move in different directions rather than uniformly favoring one model. The decline in precision at higher recall for the Benign and Malignant classes indicates that increasing the model's sensitivity to lesion regions also increases false-positive segmentation predictions, which is consistent with the background-related errors observed in the normalized confusion matrices.

F. Comparison with Related Work

The present results can be compared with several recent studies, although differences in datasets, data splits, preprocessing, training procedures, and evaluation protocols must be considered when interpreting any direct numerical comparison.

The current YOLO26s-seg configuration achieved a box mAP@50 of 0.84192 and box mAP@50:95 of 0.59732. Diphto et al. investigated lightweight YOLO models for breast ultrasound lesion segmentation using BUSI, BUS-UCLM, and BUS-BRA-related experiments and reported a YOLO26s result on BUS-BRA of approximately 0.8280 at mAP@50 and 0.6079 at mAP@50:95 [7].

Compared with those reported values, the present model obtained a higher mAP@50 but a slightly lower mAP@50:95, suggesting competitive performance under the IoU 0.50 criterion while leaving room for improvement in stricter localization quality.

Arioua et al. used a different two-stage approach on BUS-UCLM, combining YOLOv11-n localization with a dedicated segmentation architecture [3]. Their direct segmentation experiments reported Dice scores ranging from approximately 70% to 84%, depending on the architecture, with DeepLabV3 achieving the highest Dice score among their direct segmentation models. Because their study uses Dice and IoU while the present study primarily reports YOLO mAP metrics, direct numerical comparison should be treated cautiously.

Yu et al. evaluated twelve YOLO variants using same-database and cross-database experiments [6]. Their results showed that strong same-database performance could decline substantially under cross-database evaluation. This finding is particularly relevant to the present study because the current dataset combines images from multiple sources, but a dedicated external-dataset validation experiment has not yet been completed.

G. Limitations

Several limitations remain in the current study.

First, the approximately 3.2k-image dataset is primarily produced through augmentation of a smaller set of original images. Consequently, the effective diversity of unique patient cases is considerably lower than the raw augmented-image count suggests.

Second, although multiple datasets were combined, a dedicated cross-dataset generalization experiment in which one dataset is held out entirely for external testing has not yet been completed. As Yu et al. demonstrated, strong performance on a mixed dataset does not guarantee strong generalization to unseen sources [6].

Third, the current results emphasize mAP@50 and mAP@50:95. Additional class-specific metrics, Dice scores, IoU values, and per-class confusion statistics would provide a more complete analysis, particularly for distinguishing benign from malignant lesions.

Finally, deployment and CPU-based ONNX inference remain unfinished. The current model should therefore be considered an experimental research model rather than a clinically validated diagnostic system.

V. FUTURE WORK

Several improvements are planned for the next stage of this project.

First, CLAHE-based preprocessing will be evaluated to determine whether local contrast enhancement improves lesion boundary recognition. This is motivated by related breast ultrasound studies in which preprocessing and contrast enhancement have affected segmentation performance.

Second, test-time augmentation (TTA) will be investigated. The base study included in the project's literature review reported that TTA can improve detection performance, although its effect on precise delineation may differ [7]. A controlled experiment can determine whether the same behavior occurs with the present YOLO26 model.

Third, class-specific performance will be reported in more detail. Although class-level Average Precision values are reported in Section IV-E, separate evaluation of Normal, Benign, and Malignant samples using Dice and IoU would help determine whether one class is consistently more difficult for the model and whether the observed benign–malignant asymmetry persists under different metrics.

Fourth, external validation will be performed. Rather than mixing all datasets into a single training pool, one dataset will be held out entirely as an independent evaluation set. This would provide a stronger test of domain generalization, particularly

given the cross-database performance drop observed by Yu et al. [6].

Fifth, the final model will be converted to ONNX and evaluated on CPU hardware. This would allow inference speed and resource requirements to be measured outside the Colab training environment, providing a more realistic assessment of deployment feasibility.

Finally, qualitative analysis will be expanded. The current qualitative comparison in Section IV-F examines representative predictions; a more systematic review of successful predictions, false positives, false negatives, and inaccurate lesion boundaries would help identify whether errors arise from low contrast, small lesions, irregular boundaries, imaging artifacts, or differences between datasets.

VI. CONCLUSION

This study investigated the use of YOLO26 for automated breast lesion detection, segmentation, and classification from ultrasound images. The project developed from an initial BUSI-based configuration into a multi-dataset approach incorporating BUSI, BUS-UCLM, and BUS-BRA. The final balanced dataset contained 458 images per class, with approximately 3.2k images used in the principal augmented training configuration.

Multiple experiments were performed to investigate training duration, augmentation, hyperparameters, dataset size, and model size. The experiments demonstrated that simply increasing training duration did not consistently improve validation performance and that excessive or unrealistic augmentation could be counterproductive. The dataset-size experiments showed that the approximately 3.2k configuration achieved stronger box and mask mAP@50:95 results than the larger approximately 5.1k and 9.8k configurations, despite the 9.8k configuration achieving higher mAP@50 values. The augmentation strategy itself was revised across the study, moving from aggressive transformations that could produce unrealistic ultrasound appearances toward a more conservative policy emphasizing horizontal flip, limited crop, moderate rotation and shear, and controlled brightness and exposure variation.

The current YOLO26s-seg configuration achieved a box mAP@50 of 0.84192, box mAP@50:95 of 0.59732, mask mAP@50 of 0.83802, and mask mAP@50:95 of 0.55686. The model-size comparison showed that YOLO26n-seg achieved slightly higher mAP@50 values while YOLO26s-seg achieved higher mAP@50:95 values for both detection and segmentation. Class-level precision-recall analysis showed that the Normal class was consistently the best-recognized category across both models, while the Benign and Malignant classes exhibited greater precision-recall trade-offs at higher recall levels.

The findings indicate that model performance depends not only on model size or dataset quantity but also on the quality of augmentation, training configuration, and evaluation criterion. The results also reinforce findings from recent breast ultrasound research that strong performance within a dataset does not necessarily guarantee strong generalization across different datasets [6]. The current model should consequently be regarded as a research prototype intended to investigate automated breast lesion analysis rather than a clinically validated diagnostic tool. Further work on CLAHE preprocessing, test-time augmentation, external validation, and ONNX CPU inference is necessary before the system can be considered fully evaluated for deployment.

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